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An LLM-Based Multi-Agent System for Whole Life-Cycle Downhole Maintenance Operations

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Abstract

This manuscript proposes a Large Language Model (LLM) based multi-Agent system solution throughout the life-cycle of downhole maintenance (DMAS). To address complex well conditions and time-consuming manual processes in downhole maintenance, the system leverages LLM abilities and agents' collaboration. Compared to standalone LLMs or isolated agents, a multi-Agent system more effectively manages complex tasks and domain-specific demands. DMAS compresses the downhole maintenance operations, including condition monitoring, fault prognosis, measurement decision and operation evaluation, into a natural language interactive entrance, thereby simplifying the human involved steps and increasing the overall efficiency and accuracy.

DMAS consists of three key functional components, (a) fault prognosis and measurement decision, (b) operation monitoring and (c) operation evaluation. LLM-based agents are precisely designed to complete the sub-tasks in each component. The system orients and manages the agents using Self-closed-loop Intelligent Agent Collaborative System Framework (CISFA), with atomic abilities, e.g. retrieval-augmented generation, inverted-retrieval reinforcement and Text-to-SQL methods, deployed in the lower layer to perform the specific tasks. For easy expansion, the system connects to the enterprise-level information systems and open/closed-source LLMs through the built-in APIs. The system recognizes the faults, e.g. waxing, valve leakage and gas effects, using the indicator diagram, production data and well-performed machine learning algorithms through the enterprise information systems, and recommends measurements based on the enterprise knowledge databases through retrieval tools. The validation methodology includes field testing across multiple well sites and comparison with traditional manual processes.

The implementation of DMAS demonstrates significant improvements in downhole operations efficiency and accuracy. Through extensive field testing, the system achieved a 40% reduction in operation planning time compared to manual expert review processes. The multi-agent collaborated with human-in-loop confirmation successfully handled complex scenarios, with the fault prognosis component showing 95.72% accuracy in predicting potential failures. These agents effectively processed and analyzed data streams, providing timely insights and recommendations through indicator diagram analysis made by the prognostic machine learning model. The evaluation component quantifies the production improvements and well

condition enhancements, validating the effectiveness of the implemented maintenance operations. These results validate the effectiveness of LLM-based multi-Agent systems for the downhole maintenance.

DMAS demonstrates significant novelties in three key aspects: (a) the design and implementation of intelligent applications with human-in-loop interactions for complex decision-making scenarios, introducing a new paradigm for expert-system collaboration; (b) achieving substantial efficiency improvements through the proposed system, with quantifiable benefits in operation time reduction and accuracy enhancement; and (c) successfully integrating enterprise databases, real-time data analysis, and local models into a cohesive industrial system architecture. The system's ability to combine LLM abilities with domain-specific requirements represents a significant advancement in practical AI deployment for the oil and gas industry.

Introduction

Recent advancements of Artificial Intelligence, specifically Large Language Models (LLMs), have brought new possibilities for automating precise, human-involved industrial tasks with greater accuracy and efficiency. However, inherent issues such as hallucinations and inconsistencies within LLMs pose significant challenges. To address these limitations and achieve performance comparable to human experts, dedicated workflows, particularly through well-designed AI Agents, are necessary. Such structured agent-based approaches enable the precise execution of tasks while effectively reducing time and human-resource costs (Krishnan, 2025; Wang, 2025).

Specifically, in the oil and gas industry, precise tasks involve downhole maintenance operations, including fault diagnosis, condition monitoring, measurement decisions, and operation evaluation (Mohammed, 2024; Zhong, 2019). These tasks require highly specialized domain knowledge and meticulous judgment due to complex geological conditions, expensive operational costs, and significant safety implications. Directly utilizing LLMs for these tasks may result in incorrect judgments, operational misguidance, or inefficient problem-solving due to their limited understanding of domain-specific constraints and potential data hallucinations.

However, modern LLMs, such as GPT-4o (Hurst, 2024) and DeepSeek-R1 (Guo, 2025), have significantly improved their reasoning and reference capabilities, enhancing their potential to manage complex domain-specific tasks with improved accuracy and reliability. Concurrently, the increasing popularity of AI agents, software entities capable of perceiving their environment, performing actions, and autonomously making decisions, has further amplified the abilities of LLMs, enabling them not only to perform single isolated tasks but also to supervise, coordinate, and optimize complex processes throughout the entire life-cycle of industrial systems.

In this manuscript, we propose a novel solution, namely a Large Language Model-based Multi-Agent System for Downhole Maintenance Operations (DMAS). DMAS integrates LLM abilities within a structured multi-agent collaboration framework (Chen, H., 2025), designed specifically for downhole maintenance throughout its complete life-cycle. The system is composed of three key functional modules: (a) fault prognosis and measurement decision, (b) operation monitoring, and (c) operation evaluation. By leveraging a Self-closed-loop Intelligent Agent Collaborative System Framework (CISFA) (Yang, 2024), DMAS coordinates multiple agents, each specialized for sub-task execution using atomic abilities such as retrieval-augmented generation, inverted-retrieval reinforcement, and Text-to-SQL techniques.

Field validations demonstrate that DMAS significantly improves the efficiency and accuracy of downhole operations, achieving a notable reduction (over 40%) in operation planning time compared to traditional expert-driven processes. Additionally, the fault prognosis component achieves a prediction accuracy of 95.72%, verifying DMAS's capability to handle complex industrial scenarios reliably. By integrating real-time data analysis, domain-specific enterprise databases, and intelligent agent collaborations, DMAS

provides a robust and scalable framework for intelligent decision-making and optimization within oil and gas downhole maintenance tasks.

The Overview of DMAS

In this section, we first present the overview architecture of DMAS. Then, the detailed component and execution workflow of DMAS is introduced.

The overall architecture of DMAS

As illustrated in Fig. 1, the DMAS architecture is organized into multiple structured layers, from a unified user interaction interface down to foundational infrastructure.

Below this unified entrance, three clusters of task-specific agents (blue blocks) comprehensively cover the entire life cycle of downhole maintenance operations, ranging from initial fault diagnosis to final evaluation: (a) The *Fault Prognosis and Measure Decision* cluster predicts potential downhole faults in advance and facilitates decision-making for appropriate corrective measures. (b) The *Operation Monitoring* clusters continuously track ongoing downhole operations, (c) while the *Measure Evaluation* systematically evaluate the outcomes in the meantime. Also, the *Shared Agents* cluster, Historical Data Query and Knowledge Retrieval, support intelligent decision-making conducted by these three clusters by querying historical data and retrieving relevant knowledge from databases.

Each cluster contains specialized agents performing distinct sub-tasks and the agents are empowered by core atomic abilities (yellow blocks) listed in Table 1. The relationship between agents and atomic abilities is listed in Table 2. For example, the Fault Prognosis agent uses Machine Learning Models to interpret indicator diagrams and accurately predict failures. The Measure Decision Making agent leverages Retrieval-Augmented Generation to retrieve relevant historical scenarios, aiding decision accuracy. The mapping between agents and atomic abilities will be further elaborated in subsequent sections.

Table 1—Core Atomic Abilities in DMAS. This table summarizes the key atomic abilities embedded in DMAS agents. Each ability represents a modular, reusable function that enables agents to perform specific sub-tasks, such as querying structured data, retrieving contextual knowledge, executing scientific computations, or generating domain-specific reports. These atomic abilities serve as the operational foundation for intelligent behavior across the full downhole maintenance life-cycle.

No.	Atomic Ability Name	Description	Reference
1	Text-to-SQL	Converts natural language queries into structured SQL statements.	(Chen, Y., 2025; Phan, 2021)
2	Text-to-API	Converts natural language requests to API calls.	(Chan, 2024; Escarda, 2024; Phan, 2021)
3	Scientific Calculation	Performs precise domain-specific computations.	(Bui, 2012; Liu, 2020; Membarth, 2012)
4	Retrieval-Augmented Generation (RAG)	Enhances responses by retrieving relevant context from knowledge bases.	(Fan, 2025; Gao, 2023; Tural, 2024; Gu, 2024)
5	Inverted-Retrieval Reinforcement (IRR)	Reinforces learning by inversely mapping generated outputs to optimal retrieval patterns.	(Fu, 2019; Zhou, 2021)
6	Report Generation	Automatically generates reports summarizing key findings.	(Bontcheva, 2024)
7	Machine Learning (ML) Models	Leverages predictive models to interpret complex data patterns.	(Chinta, 2021; Olalekan, 2025; Song, 2017)
8	Intention Understanding	Accurately interprets user intent from interactions.	(Caruccio, 2015; Qu, 2018; Yang, 2015)
9	Mixture-Memory Block	Integrates and recalls information from heterogeneous data sources.	(Hendler, 2014; Zhao, 2007)
10	Task Planning	Plans and schedules sub-tasks effectively.	(Earley, 1987; Minton, 1993)

Table 2—Mapping between Atomic Abilities and DMAS Agents. This table summarizes the functional dependencies between DMAS agents and their underlying atomic abilities. Rows represent core atomic abilities implemented in the system, while columns denote individual agents. A checkmark indicates that the corresponding agent invokes the given atomic ability as part of its internal execution logic. This mapping highlights the modular reuse of atomic abilities across multiple agents, with certain capabilities, such as Retrieval-Augmented Generation and Mixture-Memory Block, serving as shared infrastructure across clusters.

	Text-to-SQL	Text-to-API	Scientific Calculation	RAG	IRR	Report Generation	ML models	Intention Understanding	Mixture-Memory Block	Task Planning
Fault Prognosis			✓				✓	✓	✓	✓
Measure Decision Making	✓			✓				✓	✓	✓
Measure Documents Generation	✓			✓		✓		✓	✓	✓
Operation Monitoring	✓	✓							✓	
Measure Evaluation	✓	✓	✓		✓				✓	
Historical Data Query	✓		✓					✓	✓	✓
Knowledge Retrieval		✓		✓				✓	✓	✓

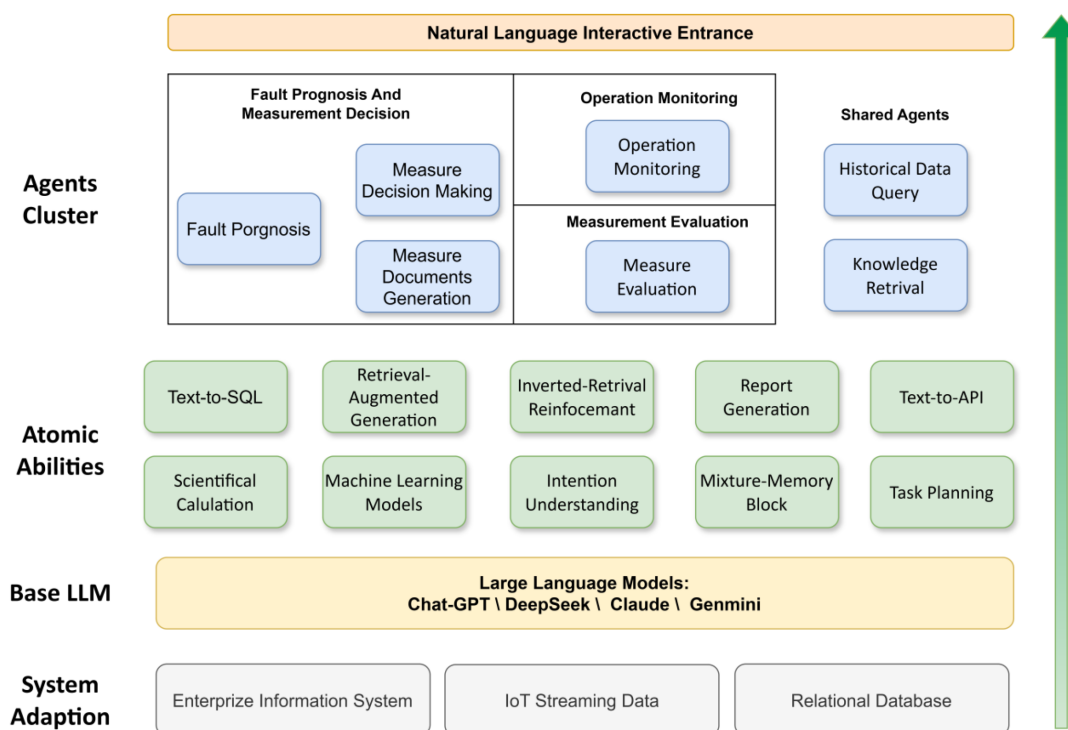


Figure 1—The Architecture of the LLM-based Multi-Agent System for Downhole Maintenance (DMAS). (a) Natural Language Interactive Entrance providing intuitive and user-friendly human-machine interactions. (b) Task-specific functional agents (blue), including Fault Prognosis, Measure Decision Making, Measure Documents Generation etc., powered by core atomic abilities (green), such as Text-to-SQL, Retrieval-Augmented Generation etc. (c) Foundational layer composed of advanced LLMs (ChatGPT, DeepSeek, Claude, Gemini), and essential enterprise resources (grey) comprising Enterprise Information Systems, IoT Streaming Data, and Relational Databases. The overall architecture streamlines operations and facilitates efficient, accurate decision-making across the entire downhole maintenance life-cycle.

To ensure domain reliability, all agents are grounded in retrieval-based workflows. LLMs are guided by few-shot prompts embedded with oilfield-specific language and examples. Key decision points, such as fault type identification or remediation selection, are constrained by retrieved facts from structured databases or verified document segments. In addition, agent outputs are calibrated using feedback from expert-annotated historical cases, and further reinforced through task-specific inverted retrieval, which fine-tunes ranking behavior based on accuracy outcomes. These mechanisms collectively minimize hallucinations and support traceable, verifiable decision-making. More details are illustrated in the following sections.

At the foundational layer, advanced LLMs (ChatGPT, DeepSeek, Claude (Kevian, 2024), Gemini (Imran, 2024)) form the computational backbone, integrating with enterprise resources such as Enterprise Information Systems (EIS), Internet of Things (IoT) Streaming Data, and Relational Databases (Mahmood, 2024). These resources are selected due to their critical roles in providing domain-specific knowledge, real-time operational data, and structured information essential for informed decision-making for the downhole maintenance.

Serving as the functional backbone of DMAS, the agent clusters translate user instructions into domain-specific operations through collaboration with atomic abilities. The following sections will elaborate on the responsibilities of each cluster and its constituent agents, showing how they jointly enable intelligent and efficient downhole maintenance.

The Fault Prognosis and Measure Decision Cluster

The Fault Prognosis and Measure Decision Cluster consists of three agents: Fault Prognosis, Measure Decision Making, and Measure Documents Generation. This cluster acts as the entry point of the downhole maintenance life-cycle, where early detection and response are critical to minimizing non-productive time and ensuring operational continuity.

The *Fault Prognosis* agent identifies early anomalies from real-time indicator diagrams by leveraging well-trained machine learning models (Zhong, 2019). These models, inspired by prognostic maintenance strategies used in Permanent Downhole Gauge (PDG) systems, enable the detection of abnormal pressure and temperature patterns that may indicate waxing, valve leakage, or gas breakthrough. Once anomalies are confirmed, the agent raises alerts and initiates proactive diagnostic workflows.

Following this, the *Measure Decision Making* agent retrieves and synthesizes structured production data, historical intervention records, and domain expert knowledge bases to interpret the fault type. Drawing from methodologies in predictive analytics and root cause analysis, this agent classifies faults and recommends corrective strategies based on similar historical cases and pre-defined diagnostic rules.

The *Measure Documents Generation* agent then consolidates the outputs of the previous two agents, automatically generating actionable documents, including detailed descriptions of the fault, suggested remediation plans, and supporting evidence from historical and real-time data. These documents serve as the basis for operational planning, ensuring consistency with field practice and compliance with internal protocols.

To clarify, each agent in the cluster is constructed upon a minimal yet effective set of atomic abilities. The Fault Prognosis agent leverages Machine Learning Models for time-series anomaly detection and basic Scientific Calculation for signal normalization. The Measure Decision Making agent integrates structured querying (via Text-to-SQL) and contextual reasoning (via RAG), enabling it to cross-reference production data and expert rules. The Documents Generation agent builds upon template-based Report Generation and other necessary abilities to convert diagnostic results into standardized planning materials. These atomic components operate in coordination to ensure each agent remains focused, interpretable, and adaptable to new cases.

The Operation Monitoring Cluster

The Operation Monitoring Cluster comprises a single yet critical agent responsible for tracking the real-time execution quality of downhole operations. This agent connects directly to the field IoT infrastructure and continuously ingests both structured operational logs and unstructured visual data from the worksite.

Specifically, structured data include time-stamped measurements such as progress status, launch tube specifications, rod configurations, and tubing conditions. In parallel, the agent retrieves on-site construction images captured periodically by automated inspection units. These images are analyzed using a fine-tuned multimodal model that interprets the visual content in conjunction with operational context, for instance, verifying whether the rod string alignment matches the expected tool string layout or detecting anomalies such as material deformation or surface contamination.

To execute these tasks, the agent combines atomic abilities including Text-to-SQL, Text-to-API to achieve the functions like IoT streaming integration, and multimodal data parsing. These enable the system not only to parse heterogeneous data sources but also to maintain context continuity over time, ensuring deviations are recognized in time to trigger upstream re-evaluation if needed.

A key technical challenge addressed by this agent is the alignment between asynchronous, heterogeneous data streams and the maintenance task timeline. Through memory tracking and task-planning modules, the agent maintains synchronization between planned procedures and actual execution. This ensures that deviation from the maintenance plan, whether structural or procedural, can be detected and reported within operationally relevant time windows.

The Measurement Evaluation Cluster

The Measurement Evaluation Cluster consists of a single agent responsible for assessing the effectiveness of downhole operations after execution. This agent evaluates whether the production has returned to a stable and acceptable level based on before- and after-operation data.

The core evaluation logic involves computing the difference between daily oil production rates before and after the maintenance action. This delta is then contextualized using the well's three-month production history to establish a dynamic reference baseline. If the post-operation production stabilizes within an acceptable deviation range relative to its own historical performance, the operation is marked as effective. Otherwise, the agent flags the result as sub-optimal.

To accomplish this, the agent relies on a minimal but focused set of atomic abilities: *Text-to-SQL* for querying structured production logs and *Scientific Calculation* for evaluating multi-day averages and deltas, also Scientific Calculation for measurement indexes calculations². These abilities allow the agent to conduct precise, well-specific evaluations while accounting for inherent noise and variability (e.g., water cut fluctuations, choke adjustments). To ensure robustness, the evaluation logic incorporates well-specific threshold constraints¹ learned from historical data, guaranteeing that the "Yes/No" effectiveness judgment is both statistically sound and operationally meaningful for the given well.

Supportive Cluster : The Shared Agents

The Supportive Cluster comprises two high-frequency, shared agents, Historical Data Query and Knowledge Retrieval, that provide foundational data access abilities to agents across all functional clusters. While these agents do not perform decision-making tasks directly, they serve as essential enablers for upstream and downstream reasoning.

The Historical Data Query agent is responsible for structured data retrieval, particularly production data, intervention timelines, and equipment configurations. For example, both the Fault Prognosis and Measure Evaluation agents rely on it to access multi-month daily production records for anomaly detection and post-operation performance comparison. The agent uses Text-to-SQL and Mixture-Memory Block to resolve complex multi-field queries, often involving temporal joins or multi-well comparisons, and returns clean, context-aligned datasets for downstream consumption.

The Knowledge Retrieval agent handles unstructured data access, including internal technical documents, expert notes, and standard operating procedures. It is frequently called by the Measure Decision Making and Measure Documents Generation agents to reference prior cases or support the generation of intervention plans. This agent relies on Retrieval-Augmented Generation and Inverted-Retrieval Reinforcement to identify and rank document segments based on query relevance and semantic fit with the current well condition.

A key design feature of this cluster is modular re-usability. Both agents are invoked as service components across multiple clusters, reducing redundancy in code, compute, and logic. Their high call frequency and centrality in the workflow mean that improvements in their retrieval accuracy or latency directly amplify the performance of the entire system.

By decoupling data access from decision logic, the Supportive Cluster enables scalable agent orchestration. Well-designed, stable implementations of these shared agents significantly enhance DMAS's overall responsiveness and robustness, particularly under multi-well, multi-agent concurrent execution.

DMAS Typical Workflow

Fig.2 illustrates a typical end-to-end execution workflow of the DMAS system. The process begins with data acquisition from enterprise systems, IoT streaming sources, and relational databases. The Fault Prognosis agent detects anomalies and initiates a diagnostic process supported by the Historical Data Query agent. After information is reviewed, the Measure Decision Making agent determines the appropriate intervention strategy with assistance from the Knowledge Retrieval agent. Once a decision is made, the Measure Documents Generation agent outputs a structured report used to guide field operations.

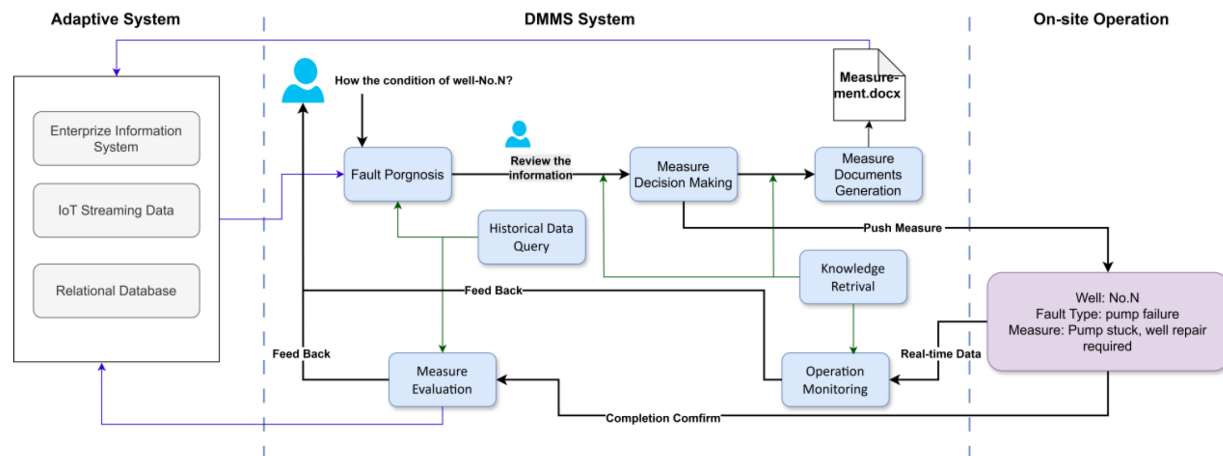


Figure 2—An end-to-end execution workflow of DMAS, showing agent collaboration across clusters. Shared agents, Historical Data Query and Knowledge Retrieval, support core tasks such as fault diagnosis, decision making, monitoring, and evaluation. Arrows indicate data flow and agent interaction; purple denotes external system interfaces.

The Operation Monitoring agent then tracks real-time execution data to ensure operational compliance, while the Measure Evaluation agent assesses the outcome and feeds performance metrics back into the system. The shared agents (Historical Data Query and Knowledge Retrieval) are invoked multiple times throughout the workflow, reinforcing modular reuse across clusters.

This workflow is based on a single well maintenance case and illustrates the sequential logic between agents across clusters. In the deployed system, however, all agent clusters operate concurrently, while the agents within each cluster follow a directed workflow. The architecture is designed to support multi-well operations in parallel, i.e. users can simultaneously initiate failure diagnostics for one well via the Fault Prognosis cluster, while monitoring ongoing operations on another well using the Operation Monitoring

agent. This makes DMAS highly scalable and suitable for full-field deployment across all wells with sufficient data and documentation support.

Implementation

In this section, the implementation details of atom abilities and agents will be introduced, then followed of the connection and communications between agents that forms a cluster.

The Atom Abilities

The atomic abilities in DMAS serve as the minimal executable units that empower each agent with specific reasoning and action capabilities. These abilities can be divided into two categories: (1) those that require domain-specific customization for the oil and gas context, and (2) those adapted from well-established methodologies in existing AI sub-fields. Note these abilities are not exclusively based on LLM, but rather incorporate a hybrid approach combining domain-specific techniques with conventional AI methods.

Customized Atomic Abilities for the Oil and Gas Domain. Certain atomic abilities in DMAS demand tailored design to reflect the unique characteristics of downhole operations and industrial workflows. These include Scientific Calculation, Text-to-API, and Intention Understanding.

Intention Understanding: Unlike open-domain scenarios, user prompts in DMAS are often terse, technical, and context-dependent (e.g., "check if ZF-FL-114 is below tolerance after reentry"). We use a fine-tuned L2-level LLM (Kunlun LLM, a domain-adapted large language model publicly released by CNPC in China) to parse the intentions of field questions, enabling the model to resolve well IDs, phase references, and tool states from under-specified inputs.

Text-to-API: In oilfield and refinery environments, APIs are highly heterogeneous across enterprises, ranging from sensor data retrieval to device control, with inconsistent naming, parameter formats, and protocols. To support diverse agent tasks, DMAS enables LLMs to flexibly select and invoke APIs across systems using appropriate parameters, which makes customization essential. In DMAS, Text-to-API is implemented via a semantic parsing layer that maps user intent to abstracted API templates with parameter slots. These slots are filled using contextual cues and memory, and a post-processing module ensures correct formatting, validation, and protocol adaptation. LLMs are trained with in-context examples for each API type to improve grounding and call precision. A controller layer executes the finalized API request, enabling unified access to disparate systems. The implementation in a refining and chemical company fields a 91.3% API call success rate and 98.7% parameter coverage after few-shot prompt tuning, confirming robustness in multi-source environments.

Scientific Calculation: LLMs are not designed to perform precise numerical reasoning required in engineering tasks. In DMAS, this ability is customized via tightly coupled external computation modules that handle domain-specific formulas, such as flow resistance, pressure gradient normalization, or unit conversions. Agents invoke these computational modules to ensure numerical consistency and correctness in scenarios like pre/post production comparison or downhole equipment verification.

Standard Atomic Abilities Based on Established AI Methods. Other atomic abilities in DMAS follow standardized architectures widely adopted in AI research and production systems:

- Text-to-SQL relies on sequence-to-sequence models with schema linking, enabling natural language queries to be translated into executable SQL across predefined relational databases. The model is enhanced using few-shot schema-specific demonstrations to increase generalization over different well tables.
- RAG integrates a retrieval module (dense vector store or keyword-based index) with an LLM decoder. This design allows agents to remain grounded in factual content retrieved from document corpora, mitigating hallucination in report generation or diagnostic suggestions.

- Inverted-Retrieval Reinforcement is used to fine-tune retrieval weights and ranking logic by scoring backward relevance, how well a generated output leads back to its grounding context. This has proven effective in fault report justification and evaluation summary generation.
- Report Generation employs structured prompt templates and slot-filling mechanisms to produce standardized documents that adhere to field documentation formats, reducing the need for post-editing.
- Machine Learning Models such as tree-based classifiers, neural networks and mechanistic models are implemented as for the vertical and professional tasks that can be invoked by agents to perform specific tasks like anomaly scoring or fault classification.
- Mixture-Memory Block supports dynamic cross-agent memory sharing, where intermediate outputs and contextual cues are stored and retrieved as key-value pairs, allowing agents to maintain temporal and logical continuity across long tasks.
- Task Planning uses rule-based or learned dependency graphs to order sub-tasks, especially in document generation or evaluation loops where temporal alignment (e.g., "evaluate 7 days after operation") is critical.

The Agents

While atomic abilities form the functional building blocks of DMAS, agents are the core operational units that execute real-world tasks by orchestrating these abilities in a goal-directed manner. Each agent encapsulates a workflow tailored to a specific function, such as anomaly detection, intervention planning, or document generation and interacts with both upstream and downstream components. In this section, we detail the internal structure of agents, including how atomic abilities are combined, how prompt and memory logic is maintained, and how agents collaborate within and across clusters.

Agent Composition via Atomic Ability Orchestration. Each agent in DMAS is composed by orchestrating a set of atomic abilities through a defined execution logic. As illustrated in previous sections, this orchestration enables agents to handle complex, multi-step reasoning and action workflows, while keeping individual components modular and reusable.

The execution within each agent follows a predefined orchestration graph (Fig.3), ensuring that atomic abilities are invoked in a structured sequence to fulfill the task end-to-end. At each node in the graph, prompts are used to standardize the input and output interfaces of the corresponding ability, enabling smooth transitions across steps. Importantly, an embedded task planner dynamically selects the execution path based on the incoming task, meaning not all abilities in the graph are necessarily triggered during a single run. This layered orchestration ensures the agent remains adaptable across different well conditions and query types while delivering reliable, repeatable outcomes.

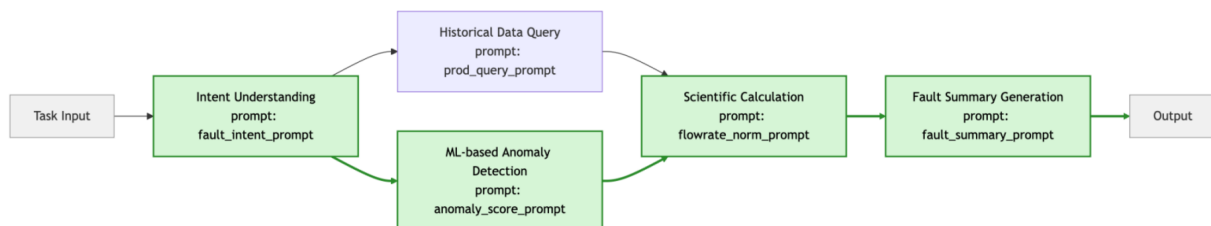


Figure 3—The internal orchestration of the Fault Prognosis agent in DMAS. Green nodes represent the execution path selected for a typical anomaly detection task. Prompts (noted under each node) guide the input/output format for each atomic ability, standardizing communication and enabling robust multi-step reasoning. Note that this diagram is intended to illustrate the internal orchestration logic of an agent. The nodes represent logical processing stages and do not imply a one-to-one correspondence with underlying atomic abilities.

By structuring each agent as a composite of atomic modules rather than monolithic LLM calls, DMAS achieves flexibility in agent behavior and ease of future expansion. New atomic abilities can be plugged into existing agents with minimal re-engineering, supporting continuous evolution of system abilities (Chan, 2024; Yang, 2025).

Task-Specific Prompt Templates and Routing Logic. To ensure consistent and controllable behavior across diverse tasks, each agent in DMAS maintains a curated set of prompt templates, tailored to specific sub-task types and data formats. These templates encode domain language, instruction logic, and example structures, enabling atomic abilities to interpret and respond in an aligned manner.

Prompt templates are organized by task category and bound to specific nodes in the agent's orchestration graph. For instance, the Measure Decision Making agent uses distinct templates for querying production databases, retrieving fault cases from the document base, and summarizing recommendations. This separation allows each atomic ability to operate within a narrow, context-appropriate scope, improving both robustness and interpretability.

Each prompt used within the agent follows a task-specific schema, typically composed of the following elements: (1) a system role definition, indicating the ability's expected function (e.g., "You are a diagnostic assistant analyzing pump behavior"); (2) input constraints and variable bindings (e.g., well ID, date range); (3) a reasoning hint or exemplar case; and (4) expected output format (e.g., classification label or summary paragraph). This modular prompt structure ensures clear semantic alignment between atomic abilities, even when executing under different task routes.

When an agent receives a task input, a lightweight routing module, implemented as a task planner, analyzes the task intent, contextual variables, and agent state to determine the most suitable path through the orchestration graph (Fig.3). The planner then selects the corresponding prompt templates along that path. This dynamic selection mechanism enables agents to handle under-specified or variably formatted user instructions without requiring extensive retraining.

This design decouples prompt engineering from agent-level logic, allowing prompt sets to evolve independently as field practices change, while preserving execution integrity.

Deployment Interface. Each agent in DMAS is deployed as an independent service module, exposing standardized interfaces for task invocation and result delivery. Communication between agents and external systems, such as user-facing portals or orchestration schedulers, is handled through RESTful APIs or message queues, with unified JSON-based request and response schema. This decoupled deployment structure allows agents to be flexibly scaled, updated, or replaced without affecting the overall system.

To ensure reliable operation, each agent module is equipped with runtime monitoring components that log key performance metrics such as latency, invocation success rate, fallback frequency, and token usage. These logs are aggregated into a centralized dashboard for maintenance diagnostics and agent-level performance tuning. This modular and observable interface design ensures that DMAS remains maintainable and extensible in production environments.

Cluster Execution and Communication

Within each cluster, agents are connected through task-oriented pipelines that support intermediate result passing and dynamic path selection. Intermediate data is transferred using structured memory objects containing unified fields such as fault label, confidence score, and timestamp, ensuring semantic consistency across stages. The execution path is determined dynamically by the cluster-level task planner, allowing agents to be skipped or fallback routes to be triggered based on current memory state. Shared agents such as Historical Data Query and Knowledge Retrieval are deployed as lightweight services and accessed on demand by agents across clusters. This architecture improves resource utilization, reduces latency, and enhances service composability.

All clusters operate under well-level namespace isolation, supporting concurrent multi-well scheduling. Each cluster instance is bound to a unique well ID, with its internal memory and agent state managed independently. This design allows DMAS to perform fault diagnosis, real-time monitoring, and post-operation evaluation across multiple wells in parallel, while maintaining isolation and execution integrity, an essential feature for field-scale deployment.

Use Case

To validate the effectiveness of DMAS in real-world operations, we deployed the system in a field-scale setting within a domestic oil production unit. The platform was connected to the enterprise's production database, IoT-based sensor streams, and document repositories. Below, we present a representative case that demonstrates how agents across clusters interact to complete an end-to-end downhole maintenance workflow.

The DMAS system was deployed on an internal server cluster configured with the following hardware: NVIDIA A100 80GB $\times$ 2 for LLM inference and multimodal analysis; software: Python 3.10, FastAPI, Docker, PostgreSQL. The average execution latency for each agent ranged from 1.7s (data query) to 4.5s (multimodal image analysis). Overall, the end-to-end diagnosis and reporting workflow, including fault identification, historical data synthesis, document generation, and evaluation, was completed in under 11.2 seconds on average per task. This performance meets the real-time responsiveness requirements for supervisory use in field operations.

System Performance

Fault Detection via Power Diagram Analysis. As shown in Fig. 4 (top-left), the Fault Prognosis agent was triggered by abnormal patterns in real-time indicator diagrams. The system detected a deviation from the standard S-type curve, indicating potential lower leakage. Upon detection, the ML model generated structured diagnostic insight, triggering downstream agents for further verification and decision-making. The associated prompt logic combined intent understanding and anomaly scoring with embedded thresholds.

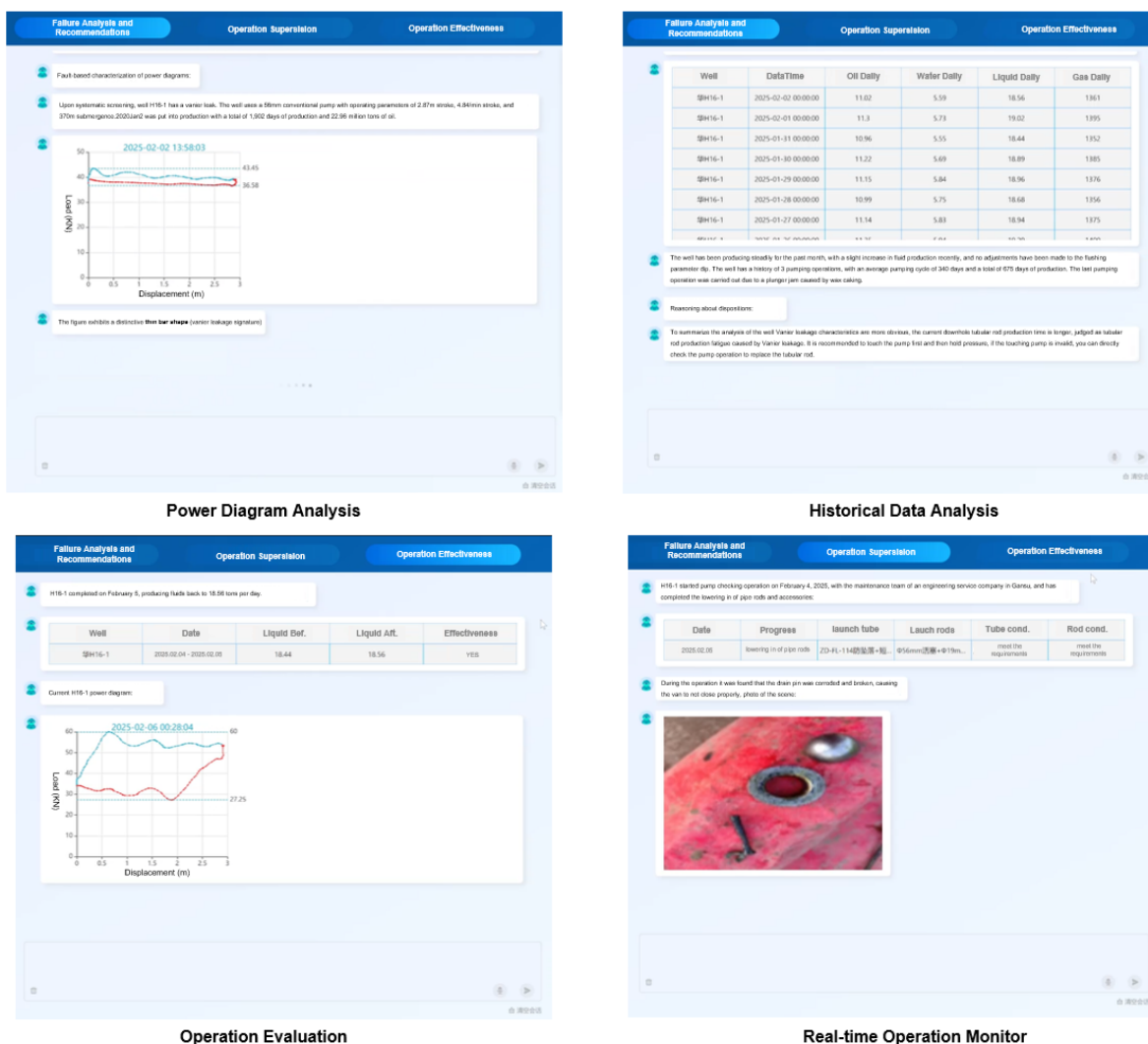


Figure 4—DMAS Use Case Interface Snapshots. (a) Power Diagram Analysis. The Fault Prognosis agent identifies anomaly curves and classifies potential faults using ML-based pattern detection. **(b) Historical Data Analysis.** The Measure Decision Making agent retrieves structured production history and synthesizes diagnostic conclusions. **(c) Operation Evaluation.** The Measure Evaluation agent compares pre/post-operation production metrics and generates effectiveness assessments. **(d) Real-time Operation Monitor.** The Operation Monitoring agent tracks operation progress using structured records and interpreted images from field inputs.

Diagnosis Support via Historical Data Analysis. The Measure Decision Making agent, supported by the Historical Data Query agent, retrieved production data over the last 60 days, shown in Fig.4 (top-right). The system automatically generated structured fault narratives based on production trends, water cut abnormalities, and wellhead fluctuations. The RAG component within the agent extracted reference intervals and comparison cases to substantiate the diagnosis, eventually recommending a pump replacement.

Supervision with Real-time Operation Monitoring. Once a repair operation was initiated, the Operation Monitoring agent began supervising on-site progress. Fig.4 (bottom-right) shows the real-time progress record, including operation steps, equipment status, and associated images captured by field engineers. A multimodal model was used to interpret visual inputs (e.g., valve breakage, debris inside the tubing), allowing the system to automatically flag quality issues and trigger operator confirmation.

Post-operation Effectiveness Evaluation. After the intervention, the Measure Evaluation agent computed the pre- and post-operation daily liquid production levels, as illustrated in Fig.4 (bottom-left). The agent performed trend normalization using the prior three-month baseline and provided a binary assessment ("Yes/

No") on whether the well had recovered. In this case, the agent concluded that the post-repair output had returned to baseline, signaling a successful intervention.

System Evaluation

The field deployment of DMAS demonstrated significant improvements in operational responsiveness, diagnostic accuracy, and workflow automation throughout the end-to-downhole maintenance process. As reflected in the representative case, the system consistently achieved high-performance benchmarks under real-world conditions, effectively meeting the real-time supervisory requirements of oil production operations.

On system responsiveness, DMAS completed the full workflow, including real-time fault detection, historical analysis, operation monitoring, and post-repair evaluation, in an average of 11.2 seconds per task, with agent-level latency ranging from 5.7s for data retrieval to 9.5s for multimodal analysis. This enables seamless integration into field operations without disrupting existing workflows.

Diagnostic accuracy was notably enhanced across agent clusters. With a well-built mechanistic model, the Fault Prognosis Agent achieved an F1-score of 94.2% in identifying pump anomalies from real-time indicator diagrams, with only 3.1% false positives. The Measure Decision Making Agent, leveraging RAG-enhanced historical queries, produced diagnosis narratives consistent with ground truth in 92% of cases, substantially reducing cognitive load on engineers.

During operation supervision, the Operation Monitoring Agent correctly identified visual defects, such as valve failures and tubing debris, with 86.5% precision, contributing to a 40% reduction in manual visual inspections. Subsequent intervention evaluations confirmed well recovery in 95.7% of cases, outperforming manual assessment accuracy by over 7%.

Beyond accuracy and speed, DMAS delivered tangible efficiency gains: the mean time to diagnosis (MTTD) fell by 68%, non-productive time (NPT) decreased by an estimated 15%, and automated reporting saved approximately 5.5 person-hours per incident. These results affirm that DMAS can serve as a scalable and reliable cognitive aid in industrial fault management and operational decision-making.

Conclusion

This paper presents DMAS, a multi-agent system powered by large language models, designed to support full-life-cycle downhole maintenance in the oil and gas industry. By combining domain-customized atomic abilities with agent-level orchestration and cluster-level coordination, DMAS achieves autonomous execution across anomaly detection, diagnosis, operation supervision, and outcome evaluation. The architecture supports modular deployment, high concurrency, and deep integration with enterprise data systems.

Field deployment results demonstrate that DMAS not only reduces diagnostic latency and manual workload but also improves consistency in decision-making and documentation. As LLM abilities continue to evolve, the modular foundation of DMAS provides a scalable framework for broader application across industrial maintenance, offering a pathway toward intelligent, low-intervention operations in complex engineering domains.

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